

Math Class 9th Federal Board
Chapter 11 Basic Statistics
PAST Paper & Mode Paper
MCQ+ Short + Long Qs

23 marks

Long $\rightarrow 1$

MCQ $\rightarrow 3$

Short $\rightarrow 3 \Rightarrow 3 \times 4 =$

8

3

12 marks

Chapter 8, 11

Most Easiest Chapters of the
Whole Book.

"CI"

13. Class mark of interval (15 – 20) is:	15	17.5	20	35
14. The probability of an event CANNOT be:	0	0.5	1	2

①
$$\frac{15 + 20}{2} = \frac{35}{2} = 17.5$$

②
$$0 \leq P(E) \leq 1$$

15. A boy scored 3 goals in 9 matches. His relative frequency of scoring goals is:

9	3	$\frac{1}{2}$	$\frac{1}{3}$
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$$\text{Relative Frequency} = \frac{3}{9} = \frac{1}{3}$$

$$0 \leq RF \leq 1$$

LLS

$$n = 7$$

13. The median of 17, 27, 37, 47, 57, 67, 77 is:	17	77	0
14. If a fair die is rolled once, what is the probability of getting 2?	$\frac{1}{2}$	$\frac{1}{6}$	$\frac{1}{5}$
15. Which is a possible Valid relative frequency value:	-0.5	1.2	0.75

$$n = 7 \Rightarrow \left(\frac{n+1}{2} \right)^{\text{th}} \text{val} = \frac{8}{2} = 4^{\text{th}} \text{val}$$

odd

Even case

$$n = 8$$

$$\frac{\frac{8}{2} + \frac{10}{2}}{2} = \frac{4^{\text{th}} + 5^{\text{th}}}{2}$$

$\left(\frac{n}{2} + \frac{n+2}{2} \right)^{\text{th}} \text{val}$

12, 14, 16

$n=8$

2, 4, 6, 8, 10, 12, 14, 16

$\frac{8}{2} = 4^{th}$ $\frac{8+2}{2} = 5^{th}$

media = 9

14. If a fair die is rolled once, what is the probability of getting 2?	$\frac{1}{2}$	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{5}$
15. Which is a possible valid relative frequency value:	-0.5	1.2	0.75	2

Die = {1, 2, 3, 4, 5, 6}

fair die $\Rightarrow P(x) = \frac{1}{6}$

$0 \leq RF \leq 1$

✓

$11 \times 7 = 77 \Rightarrow 11 \Rightarrow 62 = 11$

$11 \times 7 = 77 \Rightarrow 13 \Rightarrow \frac{64}{10} = 6.4$

xiii.	The mean of 11 numbers is 7. One of the numbers 13 is deleted. What is the mean of the remaining 10 numbers?	7.7	6.4	6.0	5.8
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Mean = $\frac{\text{Sum}}{\text{Numbers}} \Rightarrow 7 = \frac{\text{Sum}}{11}$

$\Rightarrow \text{Sum} = 77$

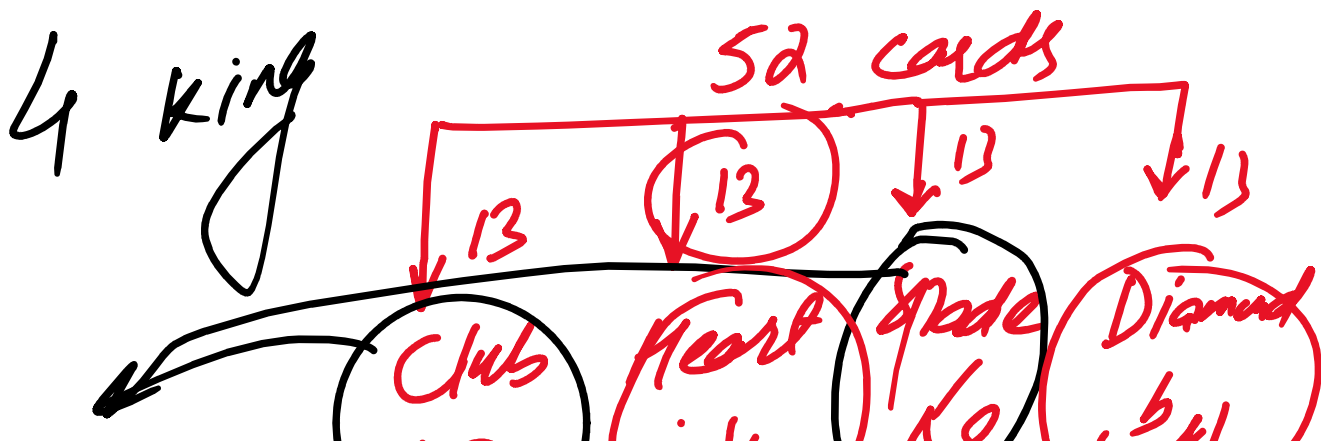
(10)

$\text{Sum} = 77 - 13 = 64$

Mean = $\frac{64}{10} = 6.4$

$\frac{4}{5} = \frac{1}{13}$

What is the probability of picking a king from well-shuffled 52 playing cards?	$\frac{1}{52}$	$\frac{1}{13}$	$\frac{4}{13}$	$\frac{1}{26}$
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Black Color

Face Cards

(K, Q, J)
 (K, Q, J)
 (K, Q, J)
 (K, Q, J)

Ace
 Ace
 Ace
 Ace

26

A fair coin is tossed twice, then the frequency of appearing head twice is:	$\frac{1}{4}$	$\frac{1}{3}$	$\frac{1}{2}$	$\frac{3}{4}$
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Coin $\rightarrow \{H, T\} \rightarrow$ Tossed once

Coin Tossed twice = $\{HH, HT, TH, TT\}$

sample space

$\frac{1}{4}$

13.	The median of the given data is: 77, 79, 82, 86, 90, 92, 93 دیے گئے مواد کا وسطانیہ کیا ہوگا؟	79	82	86	88
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$n=7 \Rightarrow \frac{7+1}{2} = 4^{th} \text{ value}$
 $\hookrightarrow \text{median}$

What is the probability of getting Tail, if a fair coin is tossed once?

اگر ایک مناسب سکہ اچھالا جائے تو 'ٹیل' آنے کا کتنا امکان ہے؟

0

$\frac{1}{2}$

1

2

$S = \{H, T\}$

$\Rightarrow P(H) = \frac{1}{2}$

$P(T) = \frac{1}{2}$

$\{1, 2, 3, 4, 5, 6\}$ $50 \div 15$

A die was rolled 50 times, and the number 4 appeared 15 times. What is the relative frequency of rolling a 4?

لڈو کا ایک دانہ 50 مرتبہ پھینکنے پر 15 مرتبہ نمبر 4 آیا۔ نمبر 4 کی ریلیٹو فریکوئنسی کیا ہوگی؟

0.15

0.30

0.20

0.25

$RF =$

$\frac{15}{50} = 0.3$

A company surveyed the number of products purchased in a month: {2, 3, 3, 2, 4, 5, 3, 6}. What is the mode of the data?

ایک کمپنی نے پراڈکٹس کی خرید کا سروے کیا۔ جس میں درج شمار سامنے آئے۔
 اس مواد کا عائدہ معلوم کریں۔ {2, 3, 3, 2, 4, 5, 3, 6}

3

4

5

6

$\text{Mode} = 3 \checkmark$

A bag contains 4 blue balls and 6 red balls. If one ball is drawn at random, what is the probability that it is **NOT** blue?

ایک بیگ میں 4 نیلی اور 6 سرخ گیندیں ہیں۔ اگر ایک گیند نکالی جائے تو اس بات کا کتنا امکان ہے کہ نکالی جانے والی گیند نیلی نہیں ہے؟

$\frac{2}{5}$

$\frac{3}{5}$

$\frac{4}{5}$

$\frac{6}{5}$

$$\text{Total Balls} = 4 + 6 = 10$$

Blue Red

$$P(\text{Not Blue}) = \frac{6}{10} = \frac{3}{5}$$

In 50 trials, an event occurs 10 times. What is the relative frequency of the event?

10 کوششوں میں ایک واقعہ 10 مرتبہ وقوع پذیر ہوتا ہے۔ اس واقعہ کی ریلیٹو فیکوئنسی کتنی ہے؟

$\frac{1}{5}$

$\frac{1}{10}$

$\frac{1}{2}$

$\frac{1}{50}$

$$RF = \frac{10}{50} = \frac{1}{5}$$

Median of 5, 7, 8, 10, 12 is:

5, 7, 8, 10, 12 کا وسطیہ (Median) کیا ہے؟

8

9

10

7

$$n=5, \frac{5+1}{2} = \frac{6}{2} = 3^{\text{rd}} \text{ value}$$

Probability of drawing a heart from a deck of 52 cards is:

52 کارڈز کے سیٹ سے ہارٹ نکالنے کا امکان کیا ہے؟

$\frac{1}{4}$

$\frac{1}{13}$

$\frac{1}{8}$

$\frac{1}{2}$

$$\frac{13}{52}$$

Relative frequency of 8 successes out of 40 trials is:

40 کوششوں میں 8 کامیابیوں کی ریلیٹو فریکوئنسی کیا ہے؟

0.1

0.2

0.4

0.8

$$RF = \frac{8}{40} = \frac{1}{5} \neq 0.2$$

- xi. A normal fair die is rolled 6000 times. The expected number of 5 is
- (a) 100 (b) 5000 (c) 1000 (d) none of these
- xii. A fair coin is tossed 500 times. Expected number of tails is
- (a) 100 (b) 250 (c) 1000 (d) none of these

$$\frac{6000}{6} \times \frac{1}{6} = \frac{1000}{6} \times \frac{1}{6} = \frac{1000}{36}$$

$$\text{Expected Number} = n \times P(E)$$

$$= 6000 \times P(5)$$

$$= 5000 \times \frac{1}{6}$$

$$= 1000$$

A fair coin is tossed 500 times. Expected number of tails is

(a) 100

(b) 250

(c) 1000

(d) none of these

$$\left. \begin{aligned} P(H) &= \frac{1}{2} \\ P(T) &= \frac{1}{2} \end{aligned} \right\}$$

$$E \cdot N = 500 \times P(T)$$

$$= 500 \times \frac{1}{2}$$

$$= 250$$

in a group of 5 people, 4 like peach juice. The expected no. of people in a population of 1200 who like peach juice is

(a) 240

(b) 600

(c) 960

(d) none of these

$$\text{Exp. Number} = 1200 \times \frac{4}{5} P(E)$$

$$= 960$$

xiv. How many times Haani should toss a fair coin if he expects to get 100 tails

(a) 100

(b) 200

(c) 1000

(d) none of these

$$\text{Exp. No.} = n \times P(E)$$

$$n \times \frac{1}{2} = 100$$

200

$$n = 200 \text{ mini TB}$$

200/1

- xv. An event which can never happen is called
 (a) sure event (b) certain event (c) possible event (d) impossible event
- xvi. Two such events whose probabilities are $\frac{1}{2}$ each are called
 (a) likely (b) equally likely (c) unlikely (d) none of these
- xvii. The probability of a certain event is
 (a) 0 (b) $\frac{1}{2}$ (c) $\frac{3}{4}$ (d) 1
- xviii. The probability of an event can take the value
 (a) 0 (b) 1 (c) $\frac{1}{2}$ (d) all of these

Impossible event $\Rightarrow P(E) = 0$

- xix. The probability of an event cannot take the value
 (a) 0 (b) 1 (c) 2 (d) all of these
- xx. The probability of an impossible event is
 (a) 0 (b) 1 (c) $\frac{1}{2}$ (d) all of these

Sec B: Short Questions

In a bag with 10 balls, there are 7 black and 3 white balls. If one ball is selected at random from the bag, calculate: a. The probability of selecting a white ball. b. The probability of selecting a black ball c. The sum of all the probability	04
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Sol: (a) $P(\text{White Ball}) = \frac{3}{10} \checkmark$

(b) $P(\text{Black Ball}) = \frac{7}{10}$

(c) Sum of all probabilities

$= P(W) + P(B)$

$= \frac{3}{10} + \frac{7}{10} = \frac{3+7}{10} = \frac{10}{10} = 1 \checkmark$

The monthly incomes (in thousand rupees) of five workers are: 20, 22, 25, 90, 23. $\checkmark$
Calculate the mean and median. Which measure better represents the income level? Give reason.

Sol. Mean $\bar{x} = \frac{\sum x}{n} = \frac{20+22+25+90+23}{5}$

$\bar{x} = \frac{180}{5} = 36$

$\boxed{\bar{x} = 36} \checkmark$ Thousand

$$\bar{x} = 36 \quad \text{Thousands}$$

Median

20, 22, 23, 25, 90

$$n=5 \Rightarrow \left(\frac{n+1}{2}\right)^{\text{th}} = \text{Mid-value}$$

3rd value will equal median

Median = 23 Thousands

Median is the better measure because it is less affected by the extreme values (Po).

In a coin toss experiment, Head is obtained 15 times in 50 trials. Calculate the relative frequencies of Head and Tails. 2+2

Sol. Relative Freq^(RF) = $\frac{\text{Observed Freq}}{\text{No. of Trials}}$

$$RF(\text{Head}) = \frac{15}{50} = \left(\frac{3}{10}\right)$$

$$RF(\text{Tail}) = \frac{50-15}{50} = \left(\frac{7}{10}\right)$$

↓ Alternate method

$$RF(\text{Tail}) = 1 - \frac{3}{10} = \left(\frac{7}{10}\right)$$

The given data shows the distribution of weights (in kg) of 30 bags of rice. Calculate mean weight of the rice bags.

C - I	05-09	10-14	15-19	20-24	25-29
f	5	7	10	4	4

04

C - I	f	x Mid-value	fx
5-9	5	7	35
10-14	7	12	84
15-19	10	17	170
20-24	4	22	88

$$x = \frac{5+9}{2} = 7$$

$$\Sigma f = n$$

	'		00
25-29	4	27	108
$\Sigma f = 30$		$\Sigma fx = 485$	

$$\bar{x} = \frac{\Sigma fx}{\Sigma f} = \frac{485}{30}$$

$$\text{Mean weight} = \bar{x} = 16.16 \text{ kg.}$$

A number is chosen at random from a set of whole numbers from 1 to 50. Calculate the probability:

a. Chosen number is not a perfect square.

b. Chosen number is perfect square.

2+2

Sol: Given numbers

1, 2, 3, 4, , 49, 50

Perfect Squares b/w 1 & 50

1, 4, 9, 16, 25, 36, 49

No. of perfect squares = 7

100 1 1 1 1 1

No. of Not-perfect squares = $50 - 7 = 43$

$$(a) P(\text{not a perfect square}) = \frac{43}{50}$$

$$(b) P(\text{perfect square}) = \frac{7}{50}$$

The populations (in thousands) of nine towns are:		
24, 18, 30, 22, 26, 26, 26, 32, 36.		
Find	Mode	1+1
	Median	+2
	Mean	

Solution 9) Mode = 26 ✓

As 26 occurs 3 times

b) Median

Arrange the data in Ascending order

18, 22, 24, 26, 26, 26, 30, 32, 36

Median write

$n = 9$ (odd) $\rightarrow$ Median will be
 $(\frac{9+1}{2})^{\text{th}}$ value = 3rd value

$$\Rightarrow \boxed{\text{Median} = 26}$$

$$(c) \quad \text{Mean} = \bar{x} = \frac{\sum x}{n}$$

$$= \frac{24 + 18 + 30 + 22 + 26 + 28 + 26 + 32 + 36}{9}$$

$$\bar{x} = \frac{240}{9} \Rightarrow \boxed{\bar{x} = 26.67}$$

Thousand people -

A decagonal die labeled

4,4,4,4, 5,5, 6, 7, 8,8 is rolled once. Find
the probability of an odd number, an
even number, and a factor of 12.

Sol: Odd-numbered faces
= 5, 5, 7

$$P(\text{odd}) = \frac{3}{10}$$

$$P(\text{Even}) = \frac{7}{10}$$

Factors of 12 = 1, 2, 3, 4, 6, 12

$$P(\text{factor of } 12) = \frac{5}{10} = \frac{1}{2}$$

R

A fair die is rolled 75 times and 5 appears up 20 times, what is the relative frequency of appearing any number up except 5.

Sol₂ $S = \{1, 2, 3, 4, 5, 6\}$

freq of other numbers except 5

$$= 75 - 20 = 55$$

$$D.F. (\text{except } 5) = \frac{55}{75} = \frac{11}{15}$$

$$(RF(\text{except } 5) = \frac{25}{75} = \frac{11}{15})$$

In a 50-over cricket match, average runs scored by Pakistani team for different sessions of the innings is given below:

The score in

01 to 10 overs: 12 runs per over, Avg ✓

11 to 35 overs: 06 runs per over, ✓

36 to 50 overs: 13 runs per over. ✓

Find average runs scored by the team in an innings.

Average runs?

Session 1 overs = 10

Session 2 " = 25

Session 3 " = 15

$$\text{Average runs} = \frac{10 \times 12 + 25 \times 6 + 15 \times 13}{50}$$

$$= \frac{120 + 150 + 195}{50}$$

$$\text{Average score} = \frac{465}{50} = 9.3 \frac{\text{runs}}{\text{over}}$$

A bag contains 5 red, 3 blue and 2 green balls. One ball is drawn at random from the bag. Calculate the probability that:

✓ ~~✗~~ ✗ ✓

A bag contains 5 red, 3 blue and 2 green balls. One ball is drawn at random from the bag. Calculate the probability that:

- ایک بیگ میں 5 سرخ، 3 نیلی اور 2 سبز گیندیں ہیں۔ امکانات معلوم کریں کہ:
- a The drawn ball is red. نکالی جانے والی گیند سرخ ہو
- b The drawn ball is NOT blue. نکالی جانے والی گیند نیلی نہ ہو

So! Total Balls = 5 + 3 + 2 = 10 ✓

$$P(\text{Red}) = \frac{5}{10} = \frac{1}{2}$$

$$P(\text{Not Blue}) = \frac{7}{10} \quad \checkmark$$

The given data shows the distribution of weights (in kg) of 30 bags of rice. Calculate mean weight of the rice bags

C-I	05 - 09	10 - 14	15 - 19	20 - 24	25 - 29
f	5	7	10	4	4

درج جدول میں چاول کے تھیلوں کو اوزان کی تقسیم دی گئی ہے۔ ان کا اوسط وزن معلوم کریں۔

A fair six-sided die is rolled 60 times. Calculate the expected frequency of

- لڈو کا ایک متناسب دانہ 60 مرتبہ پھینکا گیا۔ درج شدہ ریلیٹو فریکوئنسی معلوم کریں:
- a Rolling an even number side. ایک جفت عدد آنا
- b Rolling a prime number side. ایک مفرد عدد آنا

$$E \cdot f = n \times P(E)$$

1, 2, 3, 4, 5, 6

Sol: Even No, 2, 4, 6

$$P(\text{Even}) = \frac{3}{6} = \frac{1}{2}$$

$$\begin{aligned} \text{Exp. freq} &= n \times P(E) \\ f_{\text{Even no. side}} &= 60 \times \frac{1}{2} = 30 \end{aligned}$$

b) Prime No = 2, 3, 5

$$P(\text{Prime}) = \frac{3}{6} = \left(\frac{1}{2}\right)$$

$$\begin{aligned} \text{Exp. Freq. of} \\ \text{prime no.} &= 60 \times \frac{1}{2} \\ &= 30 \end{aligned}$$

A fair die is thrown once. Find the probability that face on the die is:

لڈو کا ایک متناسب دانہ ایک مرتبہ پھینکا گیا۔ درج شدہ کے امکانات معلوم کریں:

a A Prime number.

ایک مفرد عدد آنا

b A Multiple of 3.

عدد 3 کے اضعاف

04

$$\text{Sol. } S = \{1, 2, 3, 4, 5, 6\}$$

$$a) P(\text{Prime}) = \frac{3}{6} = \frac{1}{2}$$

$$b) \text{ Multiples of 3 are } 3, 6$$

$$P(\text{multiples of 3}) = \frac{2}{6} = \frac{1}{3}$$

The given table shows distribution of daily wages in US dollars of 50 overseas employees. Calculate average wage of the employees.

C-I	10-14	15-19	20-24	25-29	30-34
f	7	20	16	3	4

04

درج جدول میں امریکی ڈالروں میں 50 ملازمین کی تنخواہیں دی گئی ہیں۔ ان کی اوسط تنخواہ معلوم کریں۔

$$x = \frac{10 + 14 + 12}{2}$$

C-I	f	x mid-value	f x	$\bar{x} = \frac{\sum fx}{\sum f}$
10-14	7	12	84	

Do Yours

A fair coin is tossed 100 times. Calculate:

ایک متناسب سک 100 مرتبہ اچھالا گیا۔ معلوم کریں:

a Expected frequency of obtaining number of Heads.

ہیڈ آنے کی متوقع فریکوئنسی

$$P(H) = \frac{1}{2}$$

b Expected frequency of obtaining number of Tails.

ٹیل آنے کی متوقع فریکوئنسی

$$P(T) = \frac{1}{2}$$

a) Emp. freq of Heads = $100 \times \frac{1}{2} = 50$
 ب) Emp. freq - Tail = $100 \times \frac{1}{2} = 50$

Find the probability of getting a prime number when a fair die is rolled.

ایک متناسب لڈو کا دانہ (fair die) پھینکنے پر مفرد عدد (prime number) حاصل ہونے کا

04

امکان معلوم کریں۔

Sol: $S = \{1, 2, 3, 4, 5, 6\}$

Prime No. = 2, 3, 5

$P(\text{prime No.}) = \frac{3}{6} = \frac{1}{2}$

In 60 trials, event A occurred 12 times. Estimate probability and express as a fraction and decimal.

04

In 60 trials, event A occurred 12 times. Estimate probability and express as a fraction and decimal.

04

60 تجربات میں واقعہ A بارہ مرتبہ پیش آیا۔ اس کا امکان معلوم کریں اور اسے کسر

(fraction) اور اعشاریہ (decimal) میں ظاہر کریں۔

Sol:- $P(A) = \frac{12}{60} = \frac{1}{5}$ ✓
 $P(A) = 0.2$ R

Find the mean, mode and median of the data:

3, 4, 6, 6, 6, 7, 7, 8, 9.

04

درج اعداد کا عادی، اوسط اور وسطانیہ معلوم کریں:

3, 4, 6, 6, 6, 7, 7, 8, 9

Sol:-
 $\text{Mean} = \frac{\sum x}{n} = \frac{3 + 4 + 6 + 6 + 6 + 7 + 7 + 8 + 9}{9}$ ✓

mode = 6

Median :- $n = 9$
 $= \left(\frac{9+1}{2}\right)^{\text{th}} \text{val} = 5^{\text{th}}$

Median = 6

Median = 0

Special case

If $n = 10$ ✓

$$\text{Median} = \left(\frac{n}{2} + \frac{n+1}{2} \right)^{\text{th value}}$$

$$= \frac{\left(\frac{10}{2} + \frac{11}{2} \right)^{\text{th value}}}{2}$$

$$= \frac{5^{\text{th value}} + 6^{\text{th value}}}{2}$$

$$= \frac{6 + 7}{2} = \frac{13}{2} = 6.5$$